THE 13×28 CALENDAR

A Proposal for a Regular Solar Civil Calendar

Specification v1.0

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Abstract

The 13×28 Calendar is a proposed reform of the civil calendar based on a simple principle:

Make the ordinary calendar maximally regular, and isolate the unavoidable astronomical irregularity rather than distributing it throughout the ordinary months.

The system contains:

- **13 months**
- **28 days per month**
- **4 weeks per month**
- **52 ordinary weeks per year**
- **4 quarters**
- **13 weeks per quarter**
- **91 days per quarter**
- **364 ordinary days per year**
- **1 intercalary day in an ordinary year**
- **2 intercalary days in a leap year**

The ordinary calendar therefore has a completely regular structure:

7 days/week → 4 weeks/month → 28 days/month → 13 months/year → 52 weeks/year → 4 quarters/year

The additional one or two days required to keep the calendar aligned with the solar year are treated as **intercalary days outside the ordinary month, week, and quarter structure**.

The proposed month names are:

Unember, Duember, Triember, Quadember, Pentember, Hexember, September, October, November, December, Undecember, Dudecember, Tridecember.

The proposal is a **solar civil calendar**. It does not attempt to synchronize its months with the lunar cycle, and it does not seek to replace religious or traditional calendars. Those systems may continue independently and may be converted to and from the civil calendar when necessary.

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1. Purpose

The purpose of the 13×28 Calendar is to create a civil calendar whose ordinary units have consistent mathematical relationships.

The Gregorian calendar successfully maintains approximate alignment with the seasonal year, but its ordinary months have unequal lengths:

- 28 or 29 days
- 30 days
- 31 days

Consequently, months contain different numbers of days and do not correspond to a fixed number of weeks.

The 13×28 Calendar takes a different approach.

Instead of modifying ordinary months to accommodate the fractional length of the solar year, it establishes a completely regular 364-day ordinary calendar and places the remaining one or two days into an explicitly defined intercalary period.

The proposal therefore separates two problems:

1. **Organizing ordinary civil time**
2. **Maintaining long-term astronomical alignment**

The ordinary calendar handles the first.

The intercalary mechanism handles the second.

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2. Historical Context and Scope of Originality

The basic concept of a 13-month calendar containing 28 days per month is **not new**.

Moses B. Cotsworth proposed the International Fixed Calendar in 1902. His proposal also contained 13 months of 28 days, with an additional day outside the ordinary week

structure. The plan was promoted internationally and was used internally by the



Eastman Kodak Company for decades. **W** Wikipedia+2

The 13×28 Calendar should therefore **not** be presented as the invention of the general 13-month/28-day concept.

Instead, this document presents a particular specification and synthesis of the concept, including:

- the proposed month-name sequence
- Monday as the first day of every ordinary month
- four 91-day quarters
- 13 weeks per quarter
- year-end intercalary days
- the treatment of intercalary days as outside ordinary month/week/quarter identity
- a proposed transition and interoperability framework
- explicit consideration of historical, religious, technical, and legal coexistence
- a proposed approach to machine-readable implementation

The purpose is to develop and evaluate this particular configuration as a potential civil-calendar standard.

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3. Fundamental Structure

The mathematical foundation is:

13 × 28 = 364

and:

28 = 4 × 7

Therefore:

13 × 4 = 52 weeks

The ordinary calendar contains exactly:

364 days = 52 weeks

The year then receives the astronomical correction required to approximate the solar year.

Ordinary year

364 ordinary days + 1 Year Day = 365 days

Leap year

364 ordinary days + 1 Year Day + 1 Leap Day = 366 days

The intercalary days do not belong to any ordinary month, week, or quarter.

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4. The Week

The existing seven-day week is retained.

The week consists of:

1. Monday
2. Tuesday
3. Wednesday
4. Thursday
5. Friday
6. Saturday
7. Sunday

The first day of every ordinary month is **Monday**.

Thus every ordinary month has exactly:

4 Mondays
4 Tuesdays
4 Wednesdays
4 Thursdays
4 Fridays
4 Saturdays
4 Sundays

The relationship between ordinary date and weekday is therefore permanent.

For example:

- Day 1 = Monday
- Day 2 = Tuesday
- Day 8 = Monday
- Day 15 = Monday
- Day 22 = Monday
- Day 28 = Sunday

This relationship is identical in every ordinary month.

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5. The Months

Each month contains exactly 28 ordinary days.

The proposed names are:

Number	Month
1	**Unember**
2	**Duember**
3	**Triember**
4	**Quadember**
5	**Pentember**
6	**Hexember**
7	**September**
8	**October**
9	**November**
10	**December**
11	**Undecember**
12	**Dudecember**
13	**Tridecember**

The naming system is intended to make month order substantially more transparent.

The historical names September, October, November and December are derived from Latin numerical roots associated with seven, eight, nine and ten. Their numerical meanings correspond to their positions in the older Roman month sequence but no longer correspond to their positions in the modern Gregorian calendar.

The proposed calendar places them at positions 7–10:

September = 7
October = 8
November = 9
December = 10

The sequence then continues:

Undecember = 11
Dudecember = 12
Tridecember = 13

The first six names similarly provide numerical indicators:

Un → Du → Tri → Quad → Pent → Hex

The naming system is therefore intended to make the order of the months inferable rather than requiring the complete sequence to be memorized independently.

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6. The Year

The ordinary year consists of 13 complete months:

13 × 28 = 364 days

The final ordinary date is therefore:

Tridecember 28

After this comes the intercalary period.

Ordinary year

Tridecember 28 → Year Day → Unember 1

Leap year

Tridecember 28 → Year Day → Leap Day → Unember 1

The next ordinary year always begins:

Monday, Unember 1

The intercalary days are not assigned ordinary weekdays.

This preserves the ordinary seven-day structure without requiring the exceptional days to become part of a recurring seven-day sequence.

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7. Intercalary Days

The intercalary mechanism is a central feature of the proposal.

7.1 Year Day

Every year contains one **Year Day** after Tridecember 28.

Year Day:

- is not part of any month
- is not part of any quarter
- is not assigned an ordinary weekday
- is not assigned an ordinary month/day date

It is a distinct annual civil date.

7.2 Leap Day

Leap years contain an additional **Leap Day**.

Leap Day occurs immediately after Year Day and before the first day of the following ordinary year.

It is likewise:

- outside all ordinary months
- outside all ordinary quarters
- outside the ordinary weekday sequence
- separately identified

This concentrates the calendar's unavoidable irregularity into an explicitly recognized exceptional period.

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8. Leap-Year Rule

The proposal may use the established Gregorian leap-year frequency:

- A year divisible by 4 is normally a leap year.

- A century year is not a leap year unless it is divisible by 400.

Thus:

- 2000 was a leap year.
- 2100 will not be.
- 2200 will not be.
- 2300 will not be.
- 2400 will be.

This produces 97 leap years in every 400-year cycle.

The average Gregorian calendar year is therefore 365.2425 days, while the tropical year is approximately 365.2422 days. The resulting difference is extremely small. **U**USNO

The proposed calendar can use this same leap-year frequency while moving the location of the intercalation from an ordinary month to the year-end exceptional period.

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9. Why the Solar Year Requires Intercalation

The tropical year is approximately:

365.2422 days

A calendar containing exactly 365 days every year would therefore fall behind the seasons by approximately 0.2422 days each year.

The Gregorian calendar addresses this through periodic leap days. The U.S. Naval Observatory describes the Gregorian rule as producing an average year of 365.2425 days. **U**USNO

The 13×28 system does not attempt to eliminate this astronomical irregularity.

Instead, it changes how the irregularity is represented.

The ordinary civil calendar remains:

364 days

and the accumulated astronomical correction is represented through:

Year Day + periodic Leap Day

The fundamental design principle is:

Do not make ordinary months irregular merely because the solar year is irregular.

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10. Quarters

The year consists of four quarters.

Each quarter contains exactly:

91 days

because:

13 × 7 = 91

and:

91 = 3 × 28 + 7

Therefore each quarter is equivalent to:

13 complete weeks

or:

3 complete months + 1 complete week

The quarter boundaries are:

Quarter 1

Unember 1 → Quadember 7

Quarter 2

Quadember 8 → He xember 14

Quarter 3

Hexember 15 → December 21

Quarter 4

December 22 → Tridecember 28

The four quarters therefore partition the entire 364-day ordinary year without remainder.

Each quarter contains:

- 13 weeks
- 91 days
- 3 full months
- 1 additional week

The final seven days of the year are therefore part of the ordinary fourth quarter; the Year Day follows the quarter and is outside it.

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11. A Note on Quarter Boundaries

The quarters are **not defined simply as four groups of 3¼ months in everyday language**.

The formal definition is:

Each quarter is a consecutive 91-day period beginning on the first day of the relevant quarter and ending on its 13th Sunday.

This matters because a quarter does not necessarily coincide with whole-month boundaries.

This is deliberate.

The priority is to make quarters equal in duration and whole in weeks.

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12. Internal Mathematical Consistency

The central advantage of the system can be represented by a chain of exact relationships:

7 days/week

↓

4 weeks/month

↓

28 days/month

↓

13 months/year

↓

52 weeks/year

↓

4 quarters/year

↓

13 weeks/quarter

↓

91 days/quarter

↓

364 ordinary days/year

The important identities are:

28 = 4 × 7

91 = 13 × 7

364 = 13 × 28

364 = 52 × 7

364 = 4 × 91

No ordinary month requires an exception.

No ordinary quarter requires an exception.

No ordinary year requires an incomplete week.

The only exceptional units are the intercalary days required by astronomy.

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13. Fixed Weekday-Date Relationships

Because every month contains exactly four weeks, every ordinary date has a fixed weekday.

For example:

Date number	Weekday
1	Monday
2	Tuesday
3	Wednesday
4	Thursday
5	Friday
6	Saturday

Date number	Weekday
7	Sunday
8	Monday
15	Monday
22	Monday
28	Sunday

This applies to every ordinary month and every ordinary year.

Therefore:

Unember 1 is always Monday.

Pentember 15 is always Monday.

Tridecember 28 is always Sunday.

The calendar is consequently a perpetual calendar for its ordinary dates.

The International Fixed Calendar had the same broad perpetual-calendar property



because its months were also fixed at 28 days. **W** Wikipedia

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14. Advantages for Recurring Schedules

A recurring monthly event has a predictable structure.

One ordinary month always contains:

4 weeks

A two-month interval always contains:

8 weeks

A three-month interval always contains:

12 weeks

A quarter contains:

13 weeks

This makes relationships between monthly, weekly and quarterly schedules easier to calculate.

For example:

Month 4 Day 12 → Month 5 Day 12

is always exactly 28 days.

There is no need to determine whether the intervening month has 28, 30 or 31 days.

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15. Advantages for Accounting and Business

The unequal lengths of Gregorian months can complicate comparisons between monthly periods.

This issue was one of the motivations behind Cotsworth's original calendar reform efforts: he observed difficulties in comparing monthly railway statistics because different months contained different numbers of working days and weekends. **W**



Wikipedia

Under the proposed calendar:

Every ordinary month = 4 weeks

and:

Every quarter = 13 weeks

This provides equal-length ordinary accounting periods.

Businesses could therefore report:

- monthly: 4 weeks
- quarterly: 13 weeks
- annually: 52 weeks + intercalary period

The system does not automatically prove that every business would be better off using it; that would require empirical testing. Its structural advantage is that the periods are equal and predictable.

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16. Advantages for Planning

The calendar provides a simple relationship between dates and weekdays.

If an organization schedules something for:

Month X Day 10

it knows that the event will always occur on the same weekday.

Likewise, a quarterly event always falls within a quarter of exactly 13 weeks.

This may simplify:

- recurring meetings
- project planning
- production schedules
- academic timetables
- staffing schedules
- periodic reporting
- subscription periods
- maintenance schedules

Again, these are proposed practical advantages that should ultimately be tested empirically.

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17. The 13-Month Disadvantage

Thirteen is less convenient than twelve for some traditional divisions.

Twelve has several useful divisors:

12 ÷ 2 = 6

12 ÷ 3 = 4

12 ÷ 4 = 3

Thirteen is prime.

A year cannot therefore be divided into two equal whole-month halves.

Proposed solution

Use weeks for larger divisions.

The ordinary year contains:

52 weeks

Therefore:

26 weeks + 26 weeks

creates two exactly equal halves.

The system therefore does not lose the ability to divide the year evenly; it simply uses **weeks rather than whole months** for that purpose.

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18. The Intercalary-Day Disadvantage

The greatest structural novelty is also one of the greatest potential complications.

The exceptional days are not ordinary dates.

This creates questions such as:

- How should a legal deadline count them?
- How should a contract specify a date on Year Day?
- What does "one month later" mean when an interval crosses Year Day?
- How should recurring annual events be handled if they occur on an intercalary day?
- How should systems that assume every date belongs to a month respond?

Proposed solution

Give intercalary days explicit civil identities.

For example:

Year Day

Leap Day

They can be referenced directly in law and software.

A legal system could define whether a period such as "30 days" includes intercalary days; this would be a statutory convention rather than a mathematical property of the calendar.

The same principle already applies whenever legal systems define how time intervals are calculated.

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19. A Critical Date-Conversion Issue

One problem requires particular attention.

It is tempting to say:

"Take the Gregorian day-of-year and simply convert that number into the corresponding 28-day month."

That works for a particular **ordinal day**, but it does not automatically preserve a fixed Gregorian month/day anniversary across leap years.

For example, Gregorian **March 1** is:

- day 60 in a common year
- day 61 in a leap year

If the new calendar simply converts ordinal day numbers, March 1 could therefore correspond to two different 13×28 dates depending on the year.

This is a genuine issue.

Proposed distinction

The system should distinguish between:

Ordinal-date equivalence

and:

Civil-date equivalence

An event may be defined as either:

1. a particular position within the solar year, or
2. an anniversary tied to a specific civil date.

A complete conversion standard must explicitly define how each type is handled.

This should be resolved through a formal conversion specification before the calendar is adopted for legal or universal civil use.

This is one of the most important open technical questions in Version 1.0.

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20. Birthdays and Anniversaries

The same distinction applies to birthdays.

A person's birthday is normally understood as an anniversary of a **civil date**, not simply as "the Nth day of the year."

Therefore a future 13×28 calendar cannot simply convert every existing Gregorian birthday once using its ordinal day and assume the resulting date will remain equivalent in every subsequent year.

Instead, the conversion standard should establish rules for preserving the intended anniversary.

Possible approaches include:

Approach A — Civil-date preservation

Convert the original Gregorian month/day into a corresponding 13×28 date according to a defined epoch and preserve that new date permanently.

Approach B — Solar-position preservation

Preserve the event's ordinal or astronomical position within the solar year.

Approach C — Explicit transition convention

Allow governments and institutions to specify which interpretation applies during the transition.

This question should be treated as an **open design issue rather than silently assumed away**.

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21. Historical Dates

Historical records should not be rewritten to make them appear to have used the new calendar.

A historical date can retain its original calendar identity.

For example:

14 October 1066 (Julian)

can remain explicitly Julian.

A modern publication could additionally provide a converted 13×28 equivalent where useful.

The principle is:

Historical dates describe the calendar actually used by the historical source.

Calendar reform should therefore not erase historical calendar information.

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22. Religious and Traditional Calendars

The 13×28 Calendar is proposed as a **civil calendar**.

It does not need to replace:

- Hindu calendars
- Islamic calendars
- Jewish calendars
- Christian liturgical calendars
- Buddhist calendars
- other traditional or religious systems

Where a religious observance is calculated using a separate calendar, that calculation can remain in its traditional system.

The resulting observance can then be expressed in the civil calendar for administrative purposes.

This creates a coexistence model:

Traditional calendar → traditional calculation

Civil calendar → civil scheduling and administration

Conversion → communication between the systems

The fact that different calendar systems already coexist means that calendar conversion itself is not an unprecedented requirement.

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23. The Continuous-Week Question

The intercalary-day design creates one particularly significant cultural question.

The ordinary calendar contains a continuous seven-day structure.

The proposal then introduces days that are outside that structure.

Some religious traditions may regard an uninterrupted seven-day sequence as religiously significant.

For those traditions, an intercalary day outside the weekly sequence could be objectionable.

This is not a mathematical defect.

It is a question of religious practice and civil accommodation.

Possible approaches include:

- allowing religious communities to maintain their own continuous weekly reckoning
- permitting alternative legal recognition of religious weekly observance
- maintaining parallel religious and civil date representations
- further studying whether an alternative intercalation method can preserve both properties

This issue should remain explicitly open rather than being dismissed.

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24. The Lunar Question

The 28-day month should **not** be described as a lunar month.

A complete lunar phase cycle is approximately 29.5 days, so a 28-day civil month does not correspond exactly to the synodic month.

The 13×28 Calendar is therefore a **solar calendar**.

Its 28-day months are chosen because:

28 = 4 × 7

not because the Moon completes a phase cycle in exactly 28 days.

This distinction is important for scientific accuracy.

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25. Seasonal Alignment

The calendar remains connected to the seasonal year through its leap-year mechanism.

The purpose of the intercalary days is not to make the ordinary year exactly equal to the tropical year.

That is impossible using only whole days.

Instead, the system approximates the tropical year over long periods using periodic intercalation, just as the Gregorian system does. The Gregorian average year of 365.2425 days is very close to the approximately 365.2422-day tropical year. **U**USNO

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26. International and Technical Interoperability

Calendar reform would affect much more than printed calendars.

Modern systems encode dates in:

- databases
- financial systems
- operating systems
- programming libraries
- government records
- transportation systems
- contracts
- scientific datasets
- communication protocols
- scheduling systems

ISO 8601-1:2019 currently defines representations of Gregorian calendar dates for information interchange, and ISO is presently developing a successor edition. **I**



ISO+1

Therefore a serious calendar reform would require a corresponding machine-readable standard.

The 13×28 Calendar should not attempt to simply reinterpret existing Gregorian fields.

It should have an explicit calendar identifier.

For example, a future standard might define a representation conceptually similar to:

NC-YYYY-MM-DD

where **NC** identifies the new civil calendar.

The exact syntax should be determined through standards work.

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27. Software Architecture

Calendar support should ideally be treated as a calendar type rather than as a global assumption.

A robust software system should conceptually support:

- Gregorian
- 13×28
- religious calendars
- historical calendars
- other required calendar systems

and provide explicit conversion functions between them.

The principle should be:

A date is not merely a number. It is a number interpreted under a specified calendar.

This would also reduce ambiguity when historical or international data is exchanged.

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28. Transition Strategy

A calendar reform should not require an instantaneous global conversion.

A staged transition is preferable.

Stage 1 — Publication

Publish the complete specification.

Stage 2 — Independent review

Invite mathematicians, astronomers, historians, programmers, economists, legal scholars, and representatives of relevant religious communities to identify problems.

Stage 3 — Reference implementation

Create reliable conversion libraries and reference software.

Stage 4 — Voluntary experimentation

Allow organizations to use the system internally while retaining Gregorian dates externally.

Stage 5 — Dual dating

Official systems may display both calendars during a transition period.

Stage 6 — Standards development

Develop machine-readable and legal standards.

Stage 7 — Institutional adoption

Organizations that find the system useful can adopt it progressively.

Stage 8 — Potential civil adoption

If sufficient evidence and public/institutional support develop, governments can consider formal civil adoption through their normal legal processes.

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29. Dual Dating

During a transition, dates could be represented simultaneously.

For example:

Triember 17, 2048 NC
[Gregorian equivalent]

This prevents older systems from becoming immediately unusable.

It also allows people to learn the new calendar without losing access to the old one.

Dual dating would be particularly valuable for:

- legal documents
- international trade
- historical research
- software
- financial records
- transportation
- scientific publications

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30. International Coordination

International adoption would be one of the largest practical challenges.

A calendar is a coordination system.

Changing it requires coordination across jurisdictions and industries.

The relevant challenge is therefore not whether a 13×28 calendar can mathematically function.

It clearly can.

The question is whether its benefits are sufficient to justify:

- conversion costs
- software changes
- legal changes
- international coordination
- public education
- historical compatibility
- cultural adaptation

This proposal therefore does not treat adoption as an automatic consequence of mathematical elegance.

It proposes that the system first be **implemented, tested and evaluated**.

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31. Advantages Compared with the Gregorian Structure

The following advantages are structural properties of the proposed calendar.

Equal ordinary months

Every ordinary month contains exactly 28 days.

Equal weeks per month

Every ordinary month contains exactly four weeks.

Permanent weekday relationships

Every ordinary date corresponds to the same weekday.

Equal quarters

Every quarter contains exactly 91 days and 13 weeks.

Uniform ordinary accounting periods

Every ordinary month contains exactly four weeks.

No special ordinary month

There is no February-like short month.

No leap-length ordinary month

Leap years do not require an ordinary month to gain an additional day.

Simple month arithmetic

One ordinary month always equals 28 days.

Regular recurring schedules

Monthly and quarterly intervals have consistent structures.

Numerically informative month names

The proposed month sequence communicates numerical order.

Clear isolation of astronomical irregularity

The unavoidable extra day or days are explicitly identified rather than hidden inside ordinary month lengths.

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32. Disadvantages

The system also has genuine disadvantages.

32.1 Thirteen months

Thirteen does not divide evenly into two or other common whole-month subdivisions.

32.2 Intercalary days

Days outside the ordinary week/month system require new legal and software conventions.

32.3 Religious considerations

Some traditions may object to interrupting the ordinary weekly sequence.

32.4 International transition

Existing global infrastructure uses Gregorian dates extensively.

32.5 Historical compatibility

Historical calendars cannot simply be erased.

32.6 Date-conversion complexity

Fixed Gregorian month/day anniversaries require careful treatment around leap years.

32.7 Public familiarity

People already know the Gregorian calendar and would need education and adaptation.

32.8 Cultural attitudes toward 13

Some cultures associate the number 13 with bad luck.

This is a cultural consideration, not a mathematical defect.

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33. Proposed Responses to the Disadvantages

The proposal does not claim that every disadvantage disappears.

Instead:

Problem	Proposed response
13-month division	Use weeks or quarters for larger divisions
Intercalary dates	Give them explicit legal identities
Religious concerns	Permit independent religious calendars and accommodations
Existing infrastructure	Introduce dual-calendar support
Historical records	Preserve original calendar labels
Leap-year conversion	Establish a formal conversion standard
Public unfamiliarity	Education and gradual transition
Cultural resistance to 13	Address culturally rather than mathematically

The purpose of these solutions is not to pretend that transition is painless.

It is to distinguish **design problems** from **implementation problems**.

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34. The Central Design Philosophy

The proposal can be summarized in one principle:

Regularize what can be regularized; isolate what cannot.

The ordinary calendar can be made completely regular.

The solar year cannot be represented by a whole number of identical ordinary days without approximation.

Therefore:

Regular ordinary structure

isolated astronomical correction

is preferred over:

irregular ordinary structure

distributed astronomical correction

This distinction is the philosophical foundation of the 13×28 Calendar.

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35. What the System Does Not Claim

The proposal does not claim that:

- 28 days is the length of a lunar month
- the Gregorian calendar is astronomically inaccurate
- 13×28 is the first calendar to use 13 months
- every person or institution would necessarily benefit from the change
- religious calendars should be abolished
- historical dates should be rewritten
- calendar reform would be easy
- cultural resistance is irrelevant
- adoption would necessarily occur

The proposal instead makes a narrower claim:

A 13-month, 28-day solar civil calendar can provide a substantially more internally regular ordinary calendar while retaining a leap-year mechanism for solar alignment.

Whether that improvement justifies the costs of transition is an empirical, institutional and societal question.

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36. Core Mathematical Summary

The complete ordinary structure is:

7 days per week

4 weeks per month

28 days per month

13 months per year

52 weeks per year

4 quarters per year

13 weeks per quarter

91 days per quarter

364 ordinary days per year

Then:

+ 1 Year Day

and, when required:

+ 1 Leap Day

Therefore:

Ordinary year

364 + 1 = 365

Leap year

364 + 1 + 1 = 366

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37. Core Case for the System

The strongest argument for the 13×28 Calendar is its internal consistency.

1. Every ordinary month is identical.

28 days = 4 complete weeks.

2. Every ordinary date has a fixed weekday.

The same date number always corresponds to the same weekday.

3. Every ordinary year contains exactly 52 complete weeks.

52 × 7 = 364.

4. Every quarter contains exactly 13 weeks.

13 × 7 = 91.

5. Every quarter has exactly the same duration.

There is no unequal-length quarter.

6. Ordinary months never change length.

Leap years do not create a 29-day ordinary month.

7. Monthly arithmetic becomes uniform.

One ordinary month always equals 28 days.

8. Quarterly arithmetic becomes uniform.

One quarter always equals 91 days.

9. The unavoidable astronomical remainder is isolated.

Instead of changing ordinary months, the system uses explicitly identified intercalary days.

10. The month names communicate numerical order.

The sequence itself becomes informative:

Un → Du → Tri → Quad → Pent → Hex → Sept → Oct → Nov → Dec → Undec → Dudec → Tridec

11. The system remains solar.

The leap-year mechanism preserves long-term seasonal alignment.

12. Other calendars can coexist.

A civil calendar does not need to abolish religious or traditional calendars.

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38. Final Specification

The 13×28 Calendar is defined by the following principles:

1. The civil year contains 13 ordinary months.

2. Every ordinary month contains exactly 28 days.

3. Every ordinary month contains exactly four seven-day weeks.

4. Monday is the first day of every ordinary month.

5. The ordinary year contains exactly 364 days and 52 complete weeks.

6. The ordinary year is divided into four quarters of exactly 91 days and 13 weeks.

7. One Year Day follows the final ordinary day of every year.

8. A Leap Day follows Year Day in leap years.

9. Intercalary days belong to no ordinary month, quarter, or weekday.

10. The leap-year rule maintains long-term alignment with the tropical year.

11. Historical and religious calendars may continue to exist independently.

12. Calendar conversions must explicitly identify the source and destination calendar.

13. The exact treatment of civil anniversaries and legal intervals crossing intercalary days must be established by a formal conversion and legal standard before universal adoption.

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39. Proposed Month Sequence

No.	Month	Days	Weeks
1	Unember	28	4
2	Duember	28	4
3	Triember	28	4
4	Quadember	28	4
5	Pentember	28	4
6	Hexember	28	4
7	September	28	4
8	October	28	4
9	November	28	4
10	December	28	4
11	Undecember	28	4
12	Dudecember	28	4

No.	Month	Days	Weeks
13	Tridecember	28	4

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40. Closing Statement

The Gregorian calendar is not a failed astronomical system. Its leap-year mechanism provides a close approximation to the tropical year. Its irregular month structure, however, is not required by the length of the solar year.

A different design is possible.

The 13×28 Calendar proposes that the ordinary civil calendar should be built around exact relationships:

7-day weeks

4-week months

28-day months

13-month years

13-week quarters

91-day quarters

52 ordinary weeks

364 ordinary days

The remaining astronomical correction is then represented separately.

The fundamental question is therefore not:

Can the Earth be forced to have exactly 364-day years?

It cannot.

The question is:

Can the ordinary human organization of those days be made substantially more regular while preserving accurate seasonal alignment?

The 13×28 Calendar proposes that it can.

It is not presented as a finished answer to every legal, cultural, historical or technical question surrounding calendar reform.

It is presented as a **formal, testable specification**.

Its next stage should therefore not be unquestioning adoption.

Its next stage should be **criticism, implementation, conversion testing, independent analysis and refinement.**

If it survives those tests, it can progress from a calendar concept to a serious candidate for civil-calendar reform.

One-Sentence Definition

The 13×28 Calendar is a solar civil calendar of 13 identical 28-day months, four 91-day quarters, 52 ordinary weeks, and one or two year-end intercalary days, designed to maximize internal calendar regularity while preserving long-term alignment with the solar year.